

Rithvik Vukka

Data Scientist

📍 California

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SUMMARY

Data Scientist with 3 years of experience specializing in advanced machine learning techniques. Proficient in developing predictive models, and driving strategic insights. Expert in Python-based data science tools, and end-to-end model development from preprocessing to deployment. Consistently delivers measurable improvements in operational efficiency and decision-making processes.

SKILLS

Methodologies: Agile, SDLC, Waterfall

Programming Languages: Python, R, SQL

Databases: MySQL, PostgreSQL, SQL Server

Big Data Technologies: Apache Spark, Hadoop

Data Visualization Tools: Tableau, Power BI, Excel

Machine Learning Algorithms and Library: Pandas, NumPy, SciPy, Scikit-learn, Keras, TensorFlow, PyTorch, XGBoost, Random Forests, Decision Trees, SVM, Neural Networks, NLP, Large Language Models

Cloud Technologies: AWS (S3, Lambda, SageMaker), Azure (Databricks, Data Factory)

Version Control & Collaboration Tools: Git, GitHub

Soft Skills: Problem-Solving, Critical Thinking, Decision Making, Strong Communication Skills, Teamwork, Attention to Detail

EDUCATION

Master of Science in Computer Science | University of California Riverside, California

December 2024

Bachelor of Technology in Computer Science and Engineering | Amrita School of Computing, India

June 2023

PROFESSIONAL EXPERIENCE

Data Scientist (AI/ML) | KT2i, California

February 2025 - Current

- Fine-tuned a multilingual **Large Language Model (LLM)** to optimize performance in English and Spanish, ensuring improved accuracy and contextual understanding.
- Preprocessed and augmented datasets using **Python (Pandas, NumPy)** to enhance training quality.
- Evaluated chatbot responses, leading to a 15% response fluency and coherence improvement.

Data Scientist | BNY Mellon, California

October 2024 - December 2024

- Collaborated with senior data scientists and risk analysts to develop machine learning models for credit risk assessment using **Python (Scikit-learn)** and **TensorFlow**, contributing to improved prediction accuracy.
- Conducted **exploratory data analysis (EDA)** and feature engineering to support fraud detection initiatives, identifying key risk factors in transactional data.
- Developed and evaluated a prototype churn prediction model using **XGBoost, Logistic Regression**, presenting insights via Python visualizations to support retention strategies.

Data Scientist | AAPI Data, California

October 2023 - September 2024

- Developed and maintained interactive visualizations of **2020 Census** data for over 40 Asian and 34 Native Hawaiian/Pacific Islander origin groups, strengthening community access to critical demographic resources.
- Managed and updated the **State Policy Portal** platform to deliver comprehensive analysis of legislative trends affecting AAPI communities.
- Performed longitudinal analysis of **state legislative representation** patterns, including detailed examination of representation through **Asian American ethnic name** identification.
- Implemented campaign finance analysis using **public FEC records** to examine contribution patterns by race and ethnicity, utilizing machine learning models based on **Bayesian Smoothing** and **Inference Group techniques**.

Data Scientist | Vivma Software Inc, India

January 2022 - June 2023

- Refined the revenue cycle by deploying **XGBoost** and **Random Forests** in **Python**, enabling **automated anomaly detection** and reducing claims processing delays by 25%.
- Collaborated with cross-functional **stakeholders** in an Agile environment to integrate ML insights into **Power BI** dashboards, improving decision-making. Automated model deployment with **Git/GitHub** for version control and reproducibility.
- Executed a patient readmission prediction model using **TensorFlow** and **Neural Networks**, achieving high accuracy in risk assessment through **AWS SageMaker**, enabling proactive intervention strategies that minimized readmission rates.

PROJECTS

EdgeAvatar: Optimized TTS for real-time speech on Nvidia Jetson Nano, achieving 500-700ms latency. Integrated real-time STT via OpenAI API, improving reliability, reducing cloud dependency, and enhancing TTS efficiency for interactive applications.

Yelp Data Analysis for Strategic Business Insights: Analyzed Yelp datasets using SQL++ on AsterixDB, optimizing spatial queries, data unnesting, and grouping for better performance. Identified customer behavior and geographic trends, providing insights to support data-driven decision-making.

Crime Data Analysis: Analyzed Chicago Crime Dataset with Spark SQL and QGIS, integrating geospatial data for crime pattern visualization. Processed 14M+ data points, optimizing analysis time by 30% to aid law enforcement resource allocation.

PUBLICATIONS

Influential Node Identification on a Multilayered Attributed Network [🔗](#)

Applied K-Means and DBSCAN clustering to identify key nodes in a multilayered attributed network using attributes like H-Index, P-Index, Unequal P-Index, and affiliations. Visualized citation networks, enhancing research accessibility and revealing cohesive node groups.